Course Project: Magnetically Steered Cardiac Ablation

Simulating a remote-navigation catheter system end to end — from the external coils to the lesion in the heart wall

ECE · Electromagnetic Fields and Waves · Fall 2026 · Soumitra R Joy, Ph.D.
Team size 2–3 · Python · Due: end of week 14

1. Why this problem

Atrial fibrillation is the most common sustained cardiac arrhythmia, and catheter ablation is a standard treatment for it: a catheter is threaded through a vein into the heart, its tip is pressed against the inner wall, and radio-frequency current is passed through the tip to heat a small volume of muscle until it stops conducting. A ring of such lesions electrically isolates the tissue that triggers the arrhythmia.

Two engineering problems decide whether the procedure works. The first is aiming. Pushing and twisting a catheter from outside the body gives poor control deep inside a beating heart, so remote magnetic navigation systems place large coils or magnets around the patient, put a small permanent magnet in the catheter tip, and steer the tip by rotating the applied field. The second is dose. Blood conducts better than myocardium, so a tip that is barely touching the wall dumps most of its power into the blood pool and leaves a lesion too shallow to block conduction. An incomplete lesion is a common reason that an ablation has to be repeated.

Those two problems are exactly the two halves of this course. Aiming is magnetostatics: the Biot–Savart law for the coil field, and `T = m × B` for the torque that turns the tip. Dose is conduction electrostatics: Laplace's equation with a piecewise conductivity, the boundary conditions at the blood–tissue interface, and Joule's law for the power that becomes heat. You will build both, verify each against a closed-form result, and then connect them.

> **What you are being asked to decide.** Given a coil design, can the system aim the catheter accurately enough, across the working volume, to deliver a lesion of useful size — and what does that coil design cost in kilowatts and in newtons of mounting load? Your report answers that with numbers.

2. The system you will model

The chain has four links, and each one is a milestone:

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coil current → field in the chest → torque on the tip magnet → seating depth against the wall → current density in tissue → lesion
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The geometry is deliberately simple so that every stage can be checked against theory. Two coaxial coils of radius a = 350 mm sit at Helmholtz spacing around the patient. The catheter tip carries a sintered NdFeB slug of 1 mm radius and 4 mm length, giving a magnetic moment of about 13 mA·m². The ablation electrode is modelled as a conducting sphere of radius 2 mm straddling the wall surface, with myocardium ($\sigma \approx 0.54$ S/m) on one side and blood ($\sigma \approx 0.70$ S/m) on the other.

Where each result comes from in the course

Milestone	What you implement	Course result it rests on
M1 Navigation field	Biot–Savart for arbitrary current loops; field map; uniformity	Eq. (5.21) $dH = I\,dl\times\hat{R}/4\pi R^2$; Eq. (5.34) on-axis loop field
M2 Steering	Force and torque on the tip magnet as a current loop; torque balance against shaft stiffness	Eq. (5.10) $F = I\oint dl\times B$; Eq. (5.11) zero net force in a uniform field; Eq. (5.20) $T = m\times B$
M3 Coil budget	Winding resistance, Joule dissipation, force between the coils	Eq. (4.80) $R = l/\sigma A$; Eq. (4.92) $P = I^2R$; Eq. (5.41) parallel-conductor force
M4 Electrode in tissue	Finite-volume solution of $\nabla\cdot(\sigma\nabla V) = 0$; impedance; power split; lesion proxy	Laplace's equation and uniqueness; Section 4-8 boundary conditions; Eq. (4.81) R as a ratio of integrals; Eq. (4.90) Joule's law
M5 System	Chain the four together and sweep the design	All of the above

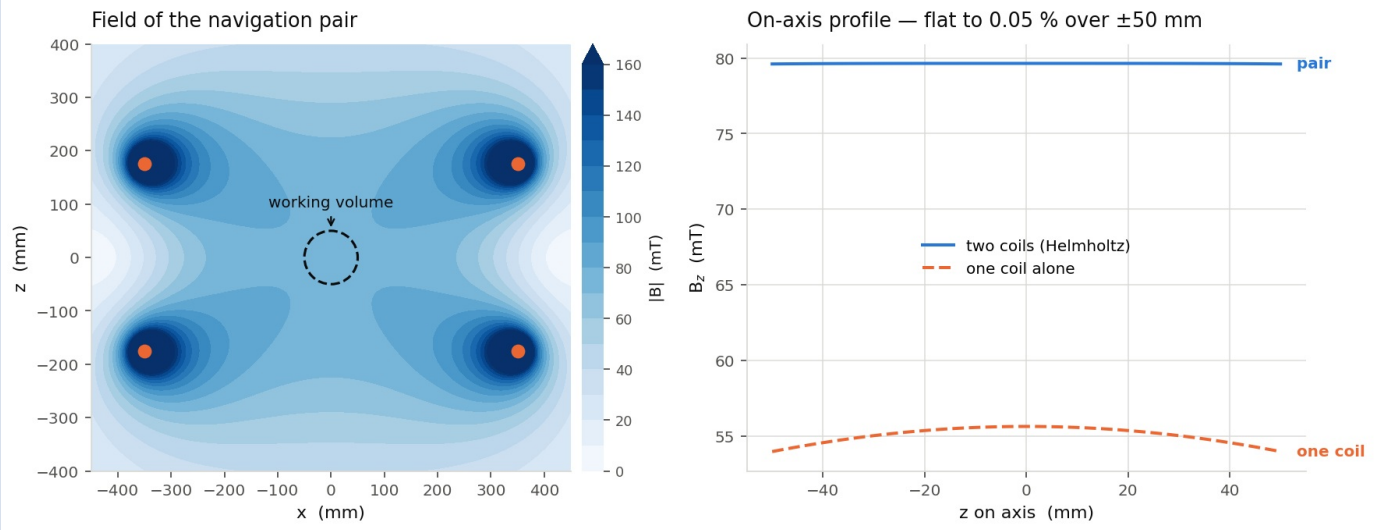
3. Milestone 1 magnetostatics

Objective. Compute the field of the navigation coils anywhere in space, and prove the routine is right.

Implement `biot_savart`, `coil_pair_field`, `on_axis_loop_bz` and `uniformity` in `emlab/magnetostatics.py`. Vectorise the Biot–Savart sum over segments; a loop-in-Python version will make Milestone 5 unpleasantly slow.

Acceptance. `pytest tests/test_m1_magnetostatics.py` must pass. It checks your numerical field against Eq. (5.34) on the axis, against $\mu_0 I/2a$ at a loop centre, against the Helmholtz centre value, and it checks that transverse components vanish on the axis by symmetry.

Report. The field map and on-axis profile, the worst-case error against theory, and the field uniformity over the ±50 mm working volume. State how many segments you needed for the error to stop improving.



Milestone 1 output. Left: |B| in the x–z plane, colour scale clipped at 160 mT so the plateau through the working volume stays readable. Right: the on-axis profile, flat to 0.05 % over ±50 mm for the pair, against a single coil carrying the same ampere-turns.

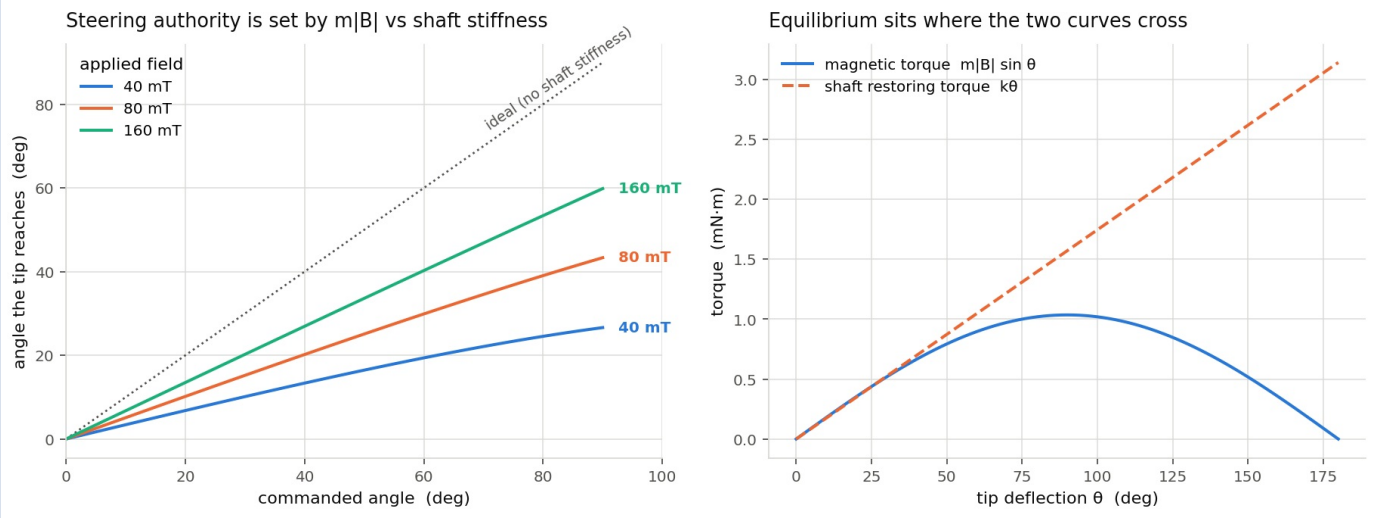
4. Milestone 2 force and torque

Objective. Find how far the applied field can actually bend the catheter tip.

Implement `moment_vector`, `force_on_loop`, `torque_on_loop` and `equilibrium_tilt` in `emlab/steering.py`. The tip magnet is represented as an equivalent current loop, so the force and torque come from the two integrals you were taught rather than from a quoted dipole formula.

Acceptance. `tests/test_m2_steering.py`. Two of its checks are physics, not bookkeeping: the net force on the loop in a uniform field must vanish (Eq. 5.11), and the loop integral for torque must agree with m×B to within a few parts in 10^3 (Eq. 5.20). If either fails, your cross products are wrong somewhere.

Report. Commanded angle versus achieved angle for at least three field strengths, and the torque-balance picture. With the baseline numbers the tip reaches only about 30° when commanded to 60°, because $m|B| \approx 1.0$ mN·m is comparable to the shaft's restoring torque. Say what you would change — field, magnet volume, or shaft stiffness — and show the calculation that supports it.



Milestone 2 output. Left: the tip undershoots the command, and the shortfall grows with angle. Right: equilibrium sits where the magnetic torque curve crosses the elastic line.

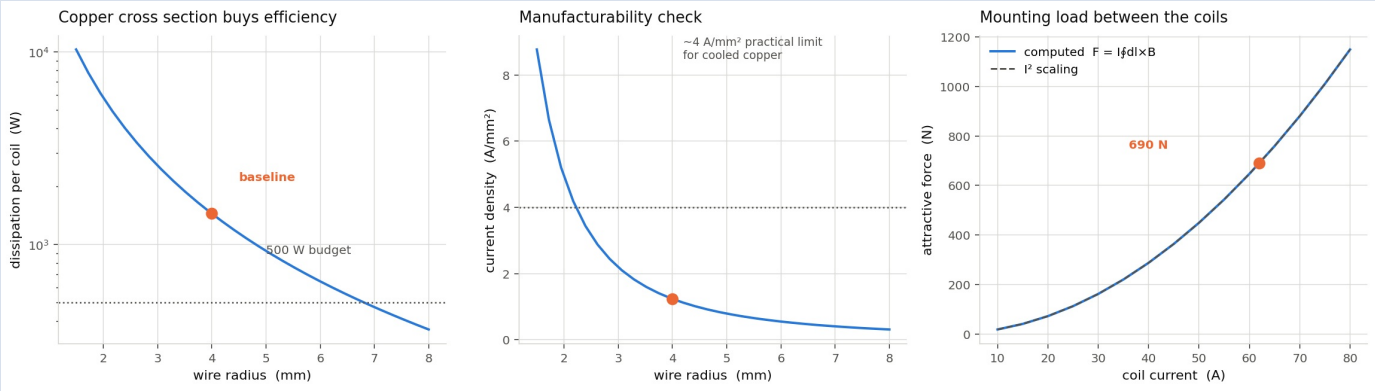
5. Milestone 3 conduction and force in the coils

Objective. Price the field you designed in Milestone 1.

Implement the functions in `emlab/thermal.py`. One of them is a trap: the field scales with the ampere-turns NI, but the heating scales with the per-turn current I squared. Using NI in $P = I^2R$ inflates the answer by a factor of N^2.

Acceptance. `tests/test_m3_thermal.py`, which includes reproducing Ulaby Exercise 4-14 ($6.9 \times 10^{-4}\ \Omega$ for 50 m of 2 cm-radius copper) and the historical ampere definition (2×10^{-7} N/m for 1 A at 1 m).

Report. The baseline design dissipates about 1.45 kW per coil and the two coils pull together with about 690 N. A lumped free-convection estimate gives a temperature rise above 250 K, which means the design as given cannot run uncooled. Carry out the wire-gauge trade study, state the cooling requirement, and give the mounting load the mechanical designer needs.



Milestone 3 output. Dissipation against conductor cross section (log scale), the current-density manufacturability check, and the attractive force between the coils, which follows I^2 as Eq. (5.41) predicts.

6. Milestone 4 conduction in tissue

Objective. Find the electrode impedance, where the RF power goes, and how large a lesion it produces.

Implement the finite-volume solver and the post-processing in `emlab/conduction.py`. The docstring gives the face-conductance recipe. Use the **harmonic** mean of σ on each face: that is what keeps the normal component of **J** continuous across the blood–myocardium interface, which is the boundary condition from Section 4-8. Try the arithmetic mean afterwards and report what changes.

Acceptance. `tests/test_m4_conduction.py`. The sharpest check is the sphere centred exactly on the interface: its field lines are radial and lie in the interface plane, so $V = A/R + B$ holds in both half spaces, each side draws current in proportion to its

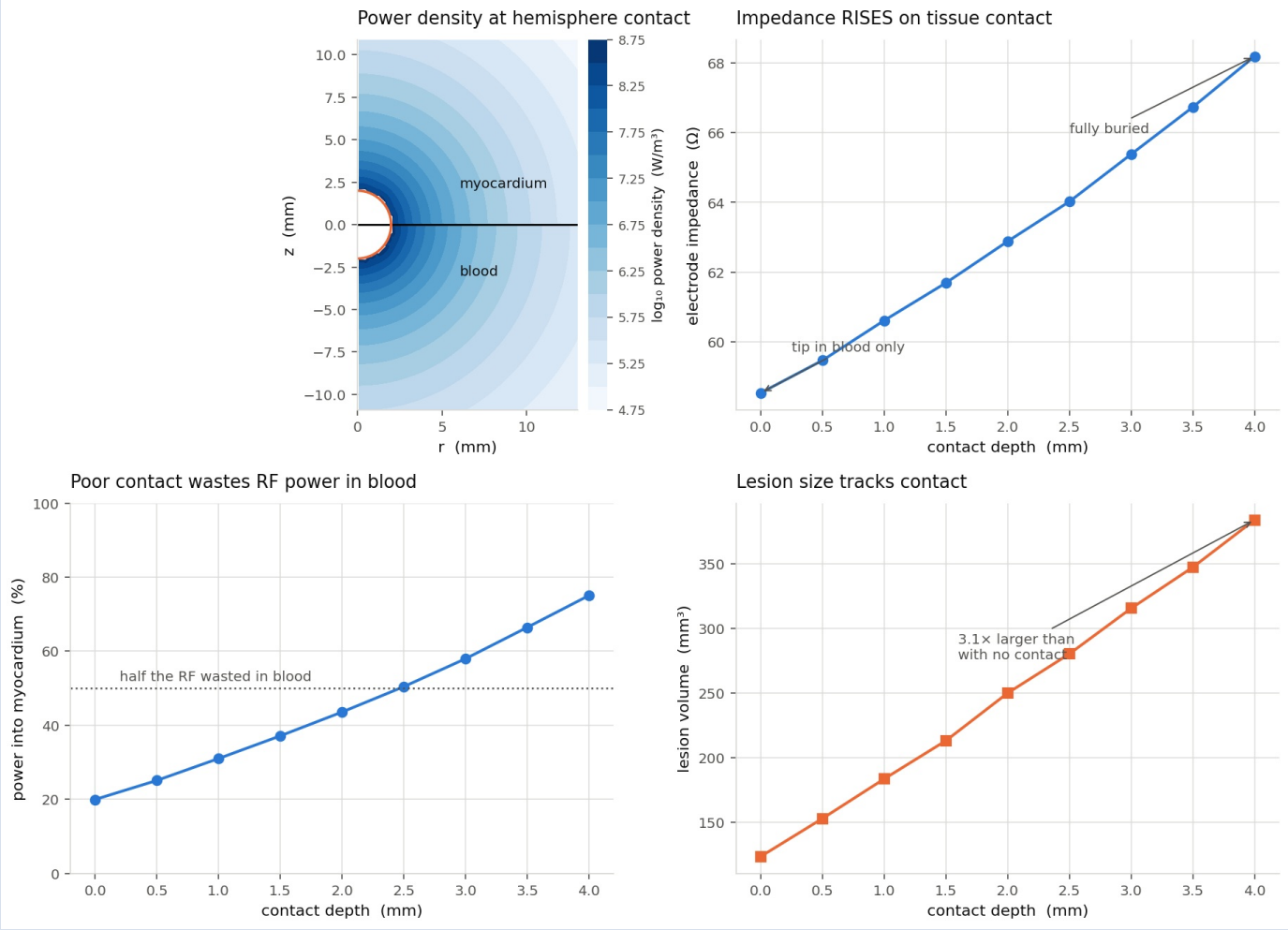
own conductivity, and

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R = (1/a − 1/R_out) / [ 2π (σ_myo + σ_blood) ]
```

Your computed power split must match σ_myo : σ_blood to about 1 %, and your resistance must approach that value as the grid is refined.

Report. A grid-refinement study. The staircase representation of a curved electrode on a Cartesian grid converges to first order — roughly 8 % error at 12 cells per tip radius, halving as the spacing halves. Report the trend, extrapolate, and state the error bar you are carrying into Milestone 5.

Then the physics: impedance against contact depth, the fraction of power reaching myocardium, and the lesion volume. The impedance *rises* on contact because myocardium conducts worse than blood — this is the signal electrophysiologists actually watch for. Explain it from the conductivity values.



Milestone 4 output. Power density around the electrode at hemisphere contact; impedance rising from 58 Ω to 68 Ω as the tip seats; the share of power reaching tissue climbing from 20 % to 75 %; and the lesion growing threefold across the same range.

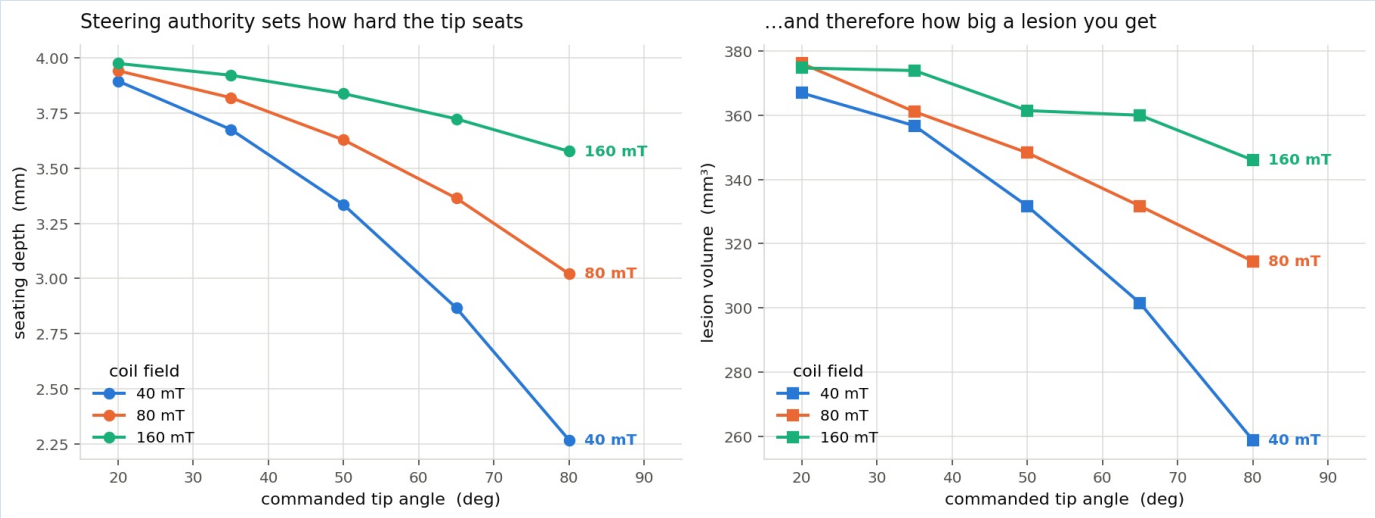
7. Milestone 5 system integration

Objective. Connect the two halves and make a design recommendation.

Chain the stages: field strength sets the steering torque, the torque balance sets the angle the tip reaches, the angular error sets how squarely the tip seats, and the seating depth sets the impedance and the lesion. Sweep commanded angle at several coil field strengths.

Report. One figure showing seating depth and lesion volume against commanded angle for at least three field levels, plus a recommendation: what field do you need so that a lesion stays above your chosen minimum volume out to the steepest angle

the procedure requires, and what does that field cost in coil dissipation and mounting force? Your answer must carry the error bar from the Milestone 4 refinement study.



Milestone 5 output. At 40 mT the lesion degrades from 367 mm³ to 259 mm³ as the commanded angle steepens, because the tip cannot be aimed well enough to seat properly. At 160 mT the same sweep stays above 346 mm³. The magnetostatics is what buys the dose.

8. The starter kit

Path	Contents
`emlab/constants.py`, `geometry.py`, `plotting.py`	Provided. Constants, the device/tissue dataclasses, and figure helpers.
`emlab/magnetostatics.py`, `steering.py`, `conduction.py`, `thermal.py`	Your work. Every function you must write raises `NotImplementedError` and carries a docstring with the governing equation and implementation hints.
`tests/`	29 acceptance tests. They are the specification — read them before you code.
`scripts/m1…m5`	Milestone drivers. Each prints the numbers for your report and writes a figure into `figures/`.

```
pip install -r requirements.txt
pytest -q # 29 failures at the start; drive it to zero
pytest tests/test_m1_magnetostatics.py -q
python scripts/m1_navigation_field.py
```

9. What to hand in

1. **Code** that passes all 29 tests, with your name(s) in the module headers.
2. **Report**, 8–12 pages: one section per milestone, each with its validation evidence, its figure, and its numbers. Every result sentence states whether the number is computed, measured, or taken from the literature.
3. **Design recommendation**, one page: the coil field you recommend, the lesion it buys, and the power and force it costs.

Limitations you are required to state

Marks are given for identifying these, and lost for claiming more than the model supports:

- The conduction model is static. RF ablation runs near 500 kHz; tissue is conduction-dominated there ($\sigma \gg \omega\varepsilon$), so the approximation is defensible, but say so and support it with numbers.
- A power-density contour is a lesion *proxy*. A real prediction needs the bioheat equation, perfusion, and a thermal-dose model. Your "lesion volume" is an iso-power surface, nothing more.
- Tissue conductivities are nominal literature values. They vary between patients, with temperature, and with direction — myocardium is anisotropic and this model treats it as isotropic.
- The coupling from steering angle to seating depth is a geometric assumption supplied in the starter kit. Critique it; a

contact-force model would be better.

- The coils are modelled as single filaments carrying NI. A real multi-layer winding has finite cross section, which changes both the near field and the inductance.

10. Grading

Weight	Criterion
30 %	Code correctness — all acceptance tests pass, and the implementations are vectorised enough to run the sweeps.
20 %	Validation evidence — each solver compared against its closed-form limit, with the error quantified and the grid-refinement study carried out.
20 %	Physical interpretation — the impedance rise, the power split, the torque balance and the coil forces explained from the governing equations.
15 %	System reasoning in Milestone 5 and the design recommendation, with the error bar carried through.
10 %	Stated limitations and honest scope.
5 %	Figure and report quality.

Extensions (bonus, up to 10 %)

- Replace the two-coil pair with a three-axis set and steer the tip to an arbitrary direction, then map steering authority over the whole working volume rather than at the centre.
- Add an irrigated electrode: hold the tip surface at a fixed temperature and solve the bioheat equation alongside the conduction problem.
- Compute the inductance of one navigation coil and estimate the supply voltage needed to slew the field in 100 ms — this reaches into Chapter 6.
- Model the catheter shaft as a flexible beam instead of a torsional spring and recover the tip trajectory, not just its final angle.